

NGUYEN LUU TAN SANG

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Summary

AI-native Full-stack Software Engineer with professional experience building web applications, internal tools, REST APIs, and data-driven features. Hands-on with TypeScript, JavaScript, Node.js, NestJS, React, Next.js, PostgreSQL, Redis, Docker, and CI/CD. Built multi-agent AI workflows using LLM orchestration, RAG, embeddings, vector search, structured validation, and deterministic fallbacks. Experienced in production debugging, automation, and shipping maintainable features with AI-assisted development.

Skills

Languages & Frameworks: TypeScript, JavaScript, Node.js, NestJS, React, Next.js

Backend & Data: REST APIs, PostgreSQL, MongoDB, Redis

AI & Automation: AI Agents, RAG, Embeddings, Vector Search

Software Engineering: OOP, SOLID Principles, Clean Code, JWT, RBAC

Testing & DevOps: Unit Testing, Integration Testing, Docker, Docker Compose, GitHub Actions, CI/CD, Git, Linux

AI-Assisted Development: Claude Code, OpenAI Codex, ChatGPT for coding, debugging, refactoring, review, and documentation

Experience

SimhiTech Technology JSC

Dec 2025 – Feb 2026

Software Engineer

- Developed internal management software for factory workforce and operations using Node.js, React, REST APIs, and MSSQL, covering workflows, dashboards, reporting, and configuration-driven features.
- Integrated Zalo Bot and Zalo Official Account (Zalo OA) with internal systems for automated operational reports and notifications, including configurable recipient and department routing.
- Investigated production issues involving APIs, database data, configurations, caching/fallback flows, and retry handling; supported Docker-based environments and CI/CD workflows.

MyAloha Technology JSC

Mar 2026 – Aug 2026

Software Engineer Intern

- Developed and enhanced 15+ features and modules for public-sector digital transformation platforms, covering workflows, reporting, data management, and role-based functionality.
- Investigated frontend-backend integration and production-data issues involving REST APIs, authorization, validation, tenant/department rules, and edge cases.
- Improved data-intensive features through database aggregation, server-side pagination and search, while reducing unnecessary duplicate API requests.

Projects

KilnFlow – AI-Assisted Manufacturing Workflow

Jul 2026 – Aug 2026

<https://github.com/sangvirgo/kilnflow>

- Built one orchestrator and five specialized AI agents for order parsing, estimation, risk/QC, scheduling, and knowledge retrieval using LLMs, RAG, embeddings, vector search, strict validation, self-correction retries, and deterministic fallbacks.

Secure Task and Document Management Platform

Jun 2026 – Aug 2026

<https://github.com/sangvirgo/task-doc-microservices>

- Built security and workflow features across ten NestJS services using PostgreSQL, Prisma, Redis, RabbitMQ, and MinIO; contributed to GitHub Actions CI/CD with tests, Docker E2E verification, GHCR images, and AWS EC2 test deployment.

Education

Posts and Telecommunications Institute of Technology

2022 – Present

Bachelor of Science in Information Technology

GPA: 3.4/4.0

Academic Scholarships: Semester 1 & 2, AY 2024–2025; Semester 1, AY 2025–2026

Certification

TOEIC Listening & Reading: 810/990